

LAB# 11

Database Connectivity

Building the Login System

In this section we'll create a login form where user can enter their username and password. When user submit the form these inputs will be verified against the credentials stored in the database, if the username and password match, the user is authorized and granted access to the site, otherwise the login attempt will be rejected.

Step 1: Creating the Login Form

Let's create a file named "login.php"

The output of the above example (i.e. login form) will look something like this:

Login

Please fill in your credentials to login.

Username

Password

Login

Don't have an account? [Sign up now.](#)

Step 2: Creating the Welcome Page

Create "welcome.php" file, where user is redirected after successful login.

If data comes from external sources like form filled in by anonymous users, there is a risk that it may contain malicious script indented to launch cross-site scripting (XSS) attacks. Therefore, you must escape this data using the PHP `htmlspecialchars()` function before displaying it in the browser, so that any HTML tag it contains becomes harmless.

For example, after escaping special characters the string `<script>alert("XSS")</script>` becomes `<script>alert("XSS")</script>` which is not executed by the browser.

Step 3: Creating the Logout Script

Now, let's create a "logout.php" file. When the user clicks on the log out or sign out link, the script inside this file destroys the session and redirect the user back to the login page.

Adding the Password Reset Feature

Finally, in this section we will add the password reset utility to our login system. Using this feature *logged in users* can instantly reset their own password for their accounts.

Let's create a file named "reset-password.php" and place the code inside it.

PHP MySQL UPDATE Query

Let's make a SQL query using the `UPDATE` statement and `WHERE` clause, after that we will execute this query through passing it to the PHP `mysqli_query()` function to update the tables records. Consider the following *persons* table inside the *demo* database:

```
-----+
| id | first_name | last_name | email          |
+-----+-----+-----+
| 1 | Peter   | Parker   | peterparker@mail.com |
| 2 | John    | Rambo    | johnrambo@mail.com   |
| 3 | Clark   | Kent     | clarkkent@mail.com   |
| 4 | John    | Carter   | johncarter@mail.com  |
| 5 | Harry   | Potter   | harrypotter@mail.com |
+-----+-----+-----+
```

The PHP code in the following example will update the email address of a person in the persons table whose *id* is equal to 1.

```
<?php
```

```
/* Attempt MySQL server connection. Assuming you are running MySQL
server with default setting (user 'root' with no password) */
```

```
$link = mysqli_connect("localhost", "root", "", "demo");
```

```
// Check connection
```

```
if($link === false){
```

```
    die("ERROR: Could not connect. " . mysqli_connect_error());
```

```
}
```

```
// Attempt update query execution
```

```
$sql = "UPDATE persons SET email='peterparker_new@mail.com' WHERE id=1";
```

```
if(mysqli_query($link, $sql)){
```

```
    echo "Records were updated successfully.";
```

```
} else {
```

```
    echo "ERROR: Could not able to execute $sql. " . mysqli_error($link);
```

```
}
```

```
// Close connection
```

```
mysqli_close($link);
```

```
?>
```

After update the persons table will look something like this:

```
+-----+-----+-----+
| id | first_name | last_name | email          |
+-----+-----+-----+
| 1 | Peter   | Parker   | peterparker_new@mail.com |
| 2 | John    | Rambo    | johnrambo@mail.com      |
| 3 | Clark   | Kent     | clarkkent@mail.com      |
| 4 | John    | Carter   | johncarter@mail.com     |
| 5 | Harry   | Potter   | harrypotter@mail.com    |
+-----+-----+-----+
```

Retrieve Data From MySQL Using PHP

For retrieve data from MySQL the SELECT statement is used. We can retrieve data from specific column or all column of a table.

To retrieve selected column data from database the SQL query is

SELECT column_name,column_name FROM table_name;

To retrieve all the column data from a table the SQL query is

SELECT * FROM table_name;

In the below example we retrieve the data from MySQL database.

In this example we used 2 file for retrieve data

- database.php- To connecting database.
- retrieve.php- For retrieve data from database

database.php

1. `<?php`
2. `$url='127.0.0.1:3306';`
3. `$username='root';`
4. `$password='';`
5. `$conn=mysqli_connect($url,$username,$password,"crud");`
6. `if(!$conn){`
7. `die('Could not Connect My Sql:'.mysql_error());`
8. `}`
9. `?>`

retrieve.php

```
<?php
include_once 'database.php';
$result = mysqli_query($conn, "SELECT * FROM myusers");
?>
<!DOCTYPE html>
<html>
<head>
<title> Retrive data</title>
</head>
<body>
<?php
if (mysqli_num_rows($result) > 0) {
?>
<table>

<tr>
<td>First Name</td>
<td>Last Name</td>
<td>City</td>
<td>Email id</td>
</tr>
<?php
$i=0;
while($row = mysqli_fetch_array($result)) {
?>
<tr>
<td><?php echo $row["first_name"]; ?></td>
<td><?php echo $row["last_name"]; ?></td>
<td><?php echo $row["city_name"]; ?></td>
<td><?php echo $row["email"]; ?></td>
</tr>
<?php
$i++;
}
?>
</table>
<?php
}
else{
    echo "No result found";
}
```

```
?>
</body>
</html>
```

Insert Data Into MySQL Using PHP

For insert data in MySQL first we have to create a table in data base.

Here we using 3 file for insert data in MySQL:

- **database.php**:For connecting data base
- **insert.php**:for getting the values from the user
- **process.php**:A PHP file that process the request

```
CREATE TABLE `employee` (
  `userid` int(8) NOT NULL,
  `first_name` varchar(55) NOT NULL,
  `last_name` varchar(55) NOT NULL,
  `city_name` varchar(55) NOT NULL,
  `email` varchar(50) NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=latin1;
```

database.php

```
<?php
$servername='localhost';
$username='root';
$password='';
$dbname = "crud";
$conn=mysqli_connect($servername,$username,$password,"$dbname");
if(!$conn){
    die('Could not Connect My Sql:' .mysql_error());
}
?>
```

insert.php

```
<!DOCTYPE html>
<html>
<body>
    <form method="post" action="process.php">
        First name:<br>
        <input type="text" name="first_name">
        <br>
```

```

        Last name:<br>
        <input type="text" name="last_name">
        <br>
        City name:<br>
        <input type="text" name="city_name">
        <br>
        Email Id:<br>
        <input type="email" name="email">
        <br><br>
        <input type="submit" name="save" value="submit">
    </form>
</body>
</html>

```

process.php

```

<?php
include_once 'database.php';
if(isset($_POST['save']))
{
    $first_name = $_POST['first_name'];
    $last_name = $_POST['last_name'];
    $city_name = $_POST['city_name'];
    $email = $_POST['email'];
    $sql = "INSERT INTO employee (first_name,last_name,city_name,email)
    VALUES ('$first_name','$last_name','$city_name','$email')";
    if (mysqli_query($conn, $sql)) {
        echo "New record created successfully !";
    } else {
        echo "Error: " . $sql . "
" . mysqli_error($conn);
    }
    mysqli_close($conn);
}
?>

```